

Phuong Uyen (Luna) Nguyen

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EDUCATION

Arizona State University — Tempe, AZ

Expected May 2027

B.S. in Computer Systems Engineering, GPA: 4.0/4.0

- Selected for Fulton Undergraduate Research Initiative (FURI), Fall 2026 (funded); Dean's List x6; New American University Award

TECHNICAL SKILLS

Languages: C, Verilog, Python, MIPS Assembly

Embedded Systems: ARM Cortex-M0+ (bare-metal, register-level), PID control, I2C, PWM/TPM, GPIO timing, FRDM-KL46Z

FPGA / RTL: Verilog, Xilinx Vivado, Nexys A7, multi-cycle MIPS processor design

ML / Computer Vision: PyTorch, TensorFlow, OpenCV, CNNs, U-Net, Roboflow annotation

Systems / Tools: Linux kernel modules, CUDA (ASU Sol supercomputer), gem5, Git, Salesforce, PeopleSoft

Hardware: EasyEDA schematic/PCB design, JLCPCB, DigiKey, multimeter/continuity testing

RESEARCH EXPERIENCE

Undergraduate Researcher, Kerner Lab (FURI) — Arizona State University

Aug 2026 – Present

Advisor: Dr. Hannah Kerner

- Conducting funded research on data-efficient agricultural field boundary segmentation using a U-Net architecture on the Fields of the World (FTW) dataset
- Coordinating project scope, literature review, and research planning with faculty advisor

Undergraduate Researcher, FURI Robotic Arm Project — Arizona State University

Jan 2025 – May 2025

- Built a computer vision pipeline achieving ~87% object detection accuracy for robotic arm manipulation tasks

WORK EXPERIENCE

Student Worker, Enrollment — EdPlus, Arizona State University

Sep 2025 – Present

- Audit and clean Salesforce records to maintain data integrity across enrollment pipelines
- Manage reporting and spreadsheets, and work in PeopleSoft to ensure enrollment data accuracy

Advising Aide, School of Manufacturing Systems and Networks — ASU Polytechnic

Sep 2024 – Aug 2025

- Provided front desk academic advising support and managed 5+ Salesforce cases daily

PROJECTS

Wall-Follower & Color-Sensing Robots — CSE 325 (Embedded Systems)

- Built FRDM-KL46Z robots with I2C color detection, TPM2 motor control, QRE1113 line sensors, and an HC-SR04 ultrasonic scanning head; debugged TPM clock conflicts, PWM configuration, and GPIO-based echo timing

Multi-Cycle MIPS Processor — CSE 320 (FPGA/Verilog)

- Implemented a multi-cycle MIPS processor in Verilog on a Nexys A7 FPGA board

Branch Predictor & Cache Simulation — CSE 420 (Computer Architecture)

- Simulated Bi-Mode, Local, and GSelect branch predictors and an LRU-IPV cache replacement policy in gem5; wrote custom CUDA kernels for ASU's Sol supercomputer

Linux Kernel Modules — CSE 330 (Operating Systems)

- Implemented a producer-consumer process monitor, an mmap-like memory allocator, and a USB block device BIO layer as Linux kernel modules

LEADERSHIP & EXTRACURRICULAR

Electrical Team Member, Desert WAVE (Women in Autonomous Vehicle Engineering)

Aug 2026 – Present

ASU RoboSub competition team, ranked top 10 globally

- Redesigning the PCB for the vehicle's hydrophone using EasyEDA and JLCPCB; support the Programming team with CV image annotation via Roboflow

Team Lead, EPICS — Arizona State University

Aug 2025 – Dec 2025

- Lead a 7-member interdisciplinary team building a real-time shelter capacity tracking system for Florida Community Innovation using barcode scanning and Firebase

Director of Outreach, CIC (prev. Outreach Associate)

Jan 2024 – May 2025

- Led a 25-member team, growing participation by 300+ members (previously managed 65+ as Outreach Associate)